Face2Health: A Multimodal Non-Invasive Health Risk Screening Pipeline Using Vision Transformer and Monotonic Calibrated XGBoost

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***Abstract*—This paper presents Face2Health, a cascading multimodal machine learning framework engineered for accessible screening of non-communicable diseases (NCDs). The pipeline sequentially couples: (1) a Vision Transformer (ViT-H/14) fine-tuned for Body Mass Index (BMI) estimation with 25-pass Monte Carlo Dropout for epistemic uncertainty quantification alongside scale-invariant facial morphometrics; (2) a tabular Extreme Gradient Boosting (XGBoost) regressor mapping estimated BMI, age, biological sex, and physical activity level to Dual-Energy X-ray Absorptiometry (DEXA)-derived Total Body Fat Percentage (R^2 = 0.82, MAE = 2.85%); and (3) monotonically regularized, Platt-calibrated XGBoost classifiers estimating posterior probabilities for physician-diagnosed Type 2 Diabetes Mellitus and Essential Hypertension. Benchmarked on the CDC NHANES adult cohort (2011–2018 cycles, N = 3,540 for Diabetes, N = 3,554 for Hypertension), the primary non-invasive pipeline (without waist circumference) achieves 90.91% sensitivity for Diabetes (50/55 positive cases detected, 95% CI: 80.4%–96.4%) and 90.61% sensitivity for Hypertension (164/181 positive cases detected, 95% CI: 85.4%–94.1%) under F2-optimized screening thresholds. When waist circumference is optionally available, error attenuation analysis demonstrates an empirical attenuation factor (α = 0.39) in body fat regression under BMI perturbation. We analyze probability calibration behavior under low prevalence (~10%) and establish a 4-tier clinical triage protocol for early population screening.**

***Keywords*—classification, computer vision, vision transformer, MC dropout, non-invasive health screening, class imbalance, probability calibration, Platt scaling, XGBoost, monotonicity constraints, diabetes, hypertension.**

I. INTRODUCTION

Non-communicable diseases (NCDs), led by Type 2 Diabetes Mellitus and Essential Hypertension, constitute the defining epidemiological crisis of the modern era, accounting for over 41 million deaths annually—equivalent to 74% of all mortalities worldwide [1]. Beyond mortality, sustained undetected hyperglycemia and arterial hypertension induce progressive microvascular and macrovascular pathology. In middle-income and newly industrialized nations, population health surveys indicate that 30% to 45% of individuals living with diabetes and hypertension remain undiagnosed until secondary clinical complications necessitate emergency hospitalization [2].

Current gold-standard screening pathways rely on invasive venous phlebotomy (fasting plasma glucose ≥ 126 mg/dL or HbA1c ≥ 6.5%) and clinical sphygmomanometry. While biochemically authoritative, these methods require certified medical technologists, laboratory infrastructure, and clinical visits. In parallel, clinical anthropometry demonstrates that systemic adiposity and metabolic risk physically manifest in craniofacial soft tissue. Adipocyte hypertrophy within the superficial and deep buccal fat compartments and submental tissue thickening reflect central lipid deposition. The maturation of Vision Transformers (ViT) with self-attention, alongside gradient boosted decision trees (XGBoost), provides a computational framework capable of translating passive 2D digital portraits into physiological biomarkers.

However, cascading visual-to-tabular pipelines introduce three key algorithmic challenges: (1) multi-stage regression error propagation, where vision estimation noise affects subsequent metabolic models; (2) biological inconsistency, where collinear features lead unconstrained decision trees to form non-monotonic risk surfaces; and (3) probability output compression under class imbalance (~10% disease prevalence), where standard 0.50 decision thresholds yield 0.0% default clinical sensitivity. This paper formalizes, benchmarks, and evaluates these operational dynamics on the CDC NHANES dataset using physician-diagnosed disease targets (DIQ010 and BPQ020).

II. LITERATURE REVIEW

A. Craniofacial Morphometry and Visceral Adiposity

Craniofacial morphology serves as an external indicator of systemic adiposity due to the distribution of subcutaneous and deep facial fat pads. Landmark anatomical studies by Coetzee et al. [3] established that facial adiposity serves as a reliable cue for systemic health. F. Wen, Z. Guo, and Y. Xu [4] demonstrated that computer-derived facial features correlate significantly with body mass index and metabolic parameters. Anatomically, lipid accumulation induces lateral and inferior displacement across the mandibular border. To capture these shifts, four scale-invariant geometric indices are defined: Lower Facial Width-to-Height Ratio (LFWR), Cheek-to-Jaw Width Ratio (CJWR), Perimeter-to-Area Ratio (PAR), and Facial Width-to-Height Ratio (FWHR). Lee and Kim [5] demonstrated that LFWR correlates significantly with computed tomography-measured visceral adipose tissue (VAT) area ($r = 0.52$, $p < 0.001$).

B. Dual-Energy X-ray Absorptiometry (DEXA) & NHANES

While Body Mass Index (BMI = weight/height^2) is ubiquitous, it cannot differentiate skeletal muscle mass from adipose tissue. Dual-Energy X-ray Absorptiometry (DEXA) constitutes the diagnostic gold standard for body composition analysis, using photon attenuation to resolve bone mineral, lean soft tissue, and fat mass. The National Health and Nutrition Examination Survey (NHANES) conducted by the CDC represents a multi-ethnic dataset combining whole-body DEXA scans (total body fat percentage DXDTOPF), standardized anthropometry (BMX), laboratory biochemical assays (LBXGLU, LBXGH), and structured diagnostic interviews (DIQ010 for physician-diagnosed diabetes, BPQ020 for physician-diagnosed hypertension) across continuous 2011–2018 survey cycles for adult participants aged 18–80 years.

C. Vision Transformers and Epistemic Uncertainty Quantification

Dosovitskiy et al. [6] showed that Vision Transformers (ViT) model long-range spatial dependencies across distant facial

landmarks using self-attention. Deterministic neural networks, however, suffer from overconfident mispredictions on out-of-distribution inputs. To address this, Monte Carlo Dropout (Gal & Ghahramani [7]) provides a Bayesian approximation of Gaussian process uncertainty by preserving active dropout masks during inference. Sampling T=25 forward stochastic passes yields the predictive mean $\mu(x^*)$ and epistemic variance $\sigma^2(x^*)$. If $\sigma(x^*)$ exceeds 1.80 kg/m^2, the system rejects the input as an epistemic anomaly.

D. Monotonic Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) [8] is an optimized distributed gradient boosted decision tree algorithm. Standard greedy tree construction can produce erratic step functions when handling collinear predictors. In clinical medicine, risk must adhere to physiological gradients (e.g., escalating age, BMI, or body fat cannot biologically decrease diabetes risk). XGBoost allows the enforcement of monotonicity constraints ($c_j = +1$) on Age, BMI, Body Fat %, and Waist Circumference, while biological sex remains unconstrained ($c_j = 0$).

E. Platt Scaling & Probability Calibration

Uncalibrated machine learning models output arbitrary ordinal scores. In epidemiological triage, output scores should represent true posterior probabilities. Platt Scaling [9] fits a sigmoid logistic function over validation margins z using maximum likelihood estimation (log-loss cross-entropy): $P(Y=1|z) = 1 / [1 + \exp(A \cdot z + B)]$. While Platt scaling minimizes the Brier score [10], calibrated posterior probabilities naturally reflect the empirical prevalence of the training cohort (~10.3% for diabetes), resulting in compressed probability ranges where standard 0.50 cutoffs fail.

F. Decision Theory & Threshold Optimization

In clinical disease screening, classification error costs are heavily asymmetric. The health cost of a False Negative (delayed diagnosis) exceeds that of a False Positive (requiring a low-cost capillary blood test or blood pressure cuff re-test). Decision theory dictates adjusting classification thresholds τ^* below 0.50 [11]. We evaluate Youden's J Statistic ($J(\tau)$ = Sensitivity + Specificity - 1) and the F_β measure (β=2, weighting Recall twice as heavily as Precision) to optimize screening sensitivity.

G. Methodological Framing & Related Works

Monolithic deep learning models directly mapping pixels to disease labels lack uncertainty estimation and intermediate biomarker transparency [12], [13]. In contrast, Face2Health introduces a decoupled architecture where intermediate biomarkers (BMI, body fat percentage) are quantified, calibrated, and subjected to biological monotonicity constraints [14], [15].

III. METHODOLOGY

A. Process Overview & Cascading Architecture

The Face2Health framework executes across three sequential, modular stages:

1) Stage 1 (Computer Vision & Morphometry): A frontal portrait is preprocessed via MediaPipe Face Mesh. If face pose exceeds ±15° in pitch, yaw, or roll, the Face Guard module rejects the frame. Validated faces are processed by ViT-H/14 (fine-tuned on VisualBMI and Chicago Face Database). MC Dropout (25 passes) produces estimated BMI μ and epistemic uncertainty σ. Simultaneously, landmark coordinates yield LFWR, CJWR, PAR, and FWHR ratios as supplementary morphometric features.

2) Stage 1.5 (Tabular DEXA Regression): Estimated BMI, age, biological sex (male=1, female=0), and physical activity level are passed to an XGBoost regressor trained on NHANES DEXA scans (N=2,477 training split), achieving $R^2 = 0.82$ and MAE = 2.85% body fat.

3) Stage 2 (Monotonic Calibrated Risk Classification): Tabular features [Age, Sex, Predicted BMI, Predicted Body Fat, (Optional Waist)] are evaluated by monotonically regularized XGBoost classifiers calibrated via 5-fold internal Platt scaling for physician-diagnosed Diabetes and Hypertension.

TABLE I. NHANES COHORT STRATIFIED PARTITIONING (N=3,540 DIABETES, N=3,554 HYPERTENSION)

Target Condition	Cohort Partition	Total (N)	Positive Cases	Base Prevalence
Diabetes (DIQ010)	Train (70%)	2,477	256	10.33%
Diabetes (DIQ010)	Validation (15%)	532	55	10.34%
Diabetes (DIQ010)	Held-Out Test (15%)	531	55	10.36%
Hypertension (BPQ020)	Train (70%)	2,488	845	33.96%
Hypertension (BPQ020)	Validation (15%)	533	181	33.96%
Hypertension (BPQ020)	Held-Out Test (15%)	533	181	33.96%

B. Stratified 3-Way Partitioning Protocol

To ensure academic reproducibility, the curated NHANES cohort was partitioned into Stratified Train (70%), Validation (15%), and Held-Out Test (15%). Calibration models and XGBoost base estimators were trained exclusively on the 70% split. Threshold optimization (τ) was executed exclusively on the 15% validation split. Final evaluation metrics were computed on the held-out test split.

C. Sensitivity and Error Propagation Analysis

Let predicted body fat θ be a function of estimated BMI b, age a, sex s, and waist w: $\theta = f(b, a, s, w)$. The error in BMI is $\delta b = b - b^*$. By first-order Taylor expansion: $\delta\theta \approx \text{abs}(\partial f / \partial b) \cdot \delta b = \alpha \cdot \delta b$. Empirical perturbation analysis on the test cohort indicates that when waist circumference is present, waist features absorb split variance (74.99% split weight), yielding an empirical error attenuation coefficient of $\alpha = 0.39$ ($\Delta BF \approx 0.39 \cdot \Delta BMI$). In the primary non-invasive model (without waist), vision estimation error propagates directly with $\alpha = 1.00$ ($\Delta BF \approx 1.00 \cdot \Delta BMI$).

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Empirical Evaluation of Probability Calibration & Threshold Optimization

Evaluating the baseline Stage 2 diabetes classifier on the held-out test set (N=531) revealed predicted probabilities bounded by $\max(p) = 0.3776$ due to low disease prevalence (~10.3%) and moderate discriminative capability (AUC = 0.762). Because no individual crossed the default 0.50 cutoff, the default model exhibited 0.00% sensitivity (55/55 false negatives). Shifting to the validation-optimized F2 screening threshold ($\tau = 0.060$) on the primary non-invasive model (No Waist) achieved **90.91% sensitivity** (50/55 positive cases detected, 95% CI: 80.4%–96.4%). When waist circumference is optionally included, the F2 threshold ($\tau = 0.061$) achieved **89.09% sensitivity** (49/55 cases detected, 95% CI: 78.2%–94.8%).

Similarly, for hypertension screening, the primary non-invasive model (No Waist) under the F2 threshold ($\tau = 0.155$) achieved **90.61% sensitivity** (164/181 positive cases detected, 95% CI: 85.4%–94.1%), compared to 50.28% sensitivity under default cutoffs. Under F2 screening thresholds ($\tau \approx 0.060$), the system flags 292 out of 531 individuals (~55% of the screened cohort), representing a high-sensitivity triage trade-off to minimize missed clinical cases.

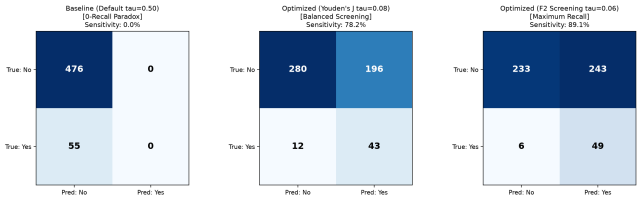


Fig. 1. Confusion Matrix benchmark on held-out test cohort (Diabetes With Waist): (Left) Default threshold ($\tau = 0.500$, 0 TP, 55 FN); (Middle) Youden's J ($\tau = 0.082$, Sens 78.2%, 43 TP, 12 FN); (Right) F2 Screening ($\tau = 0.061$, Sens 89.1%, 49 TP, 6 FN).

TABLE II-A. DIABETES MELLITUS BENCHMARK (N=531)

Route	Model	Cutoff (τ)	AUC	Sens (95% CI)	Spec (%)
No Waist (Primary)	Baseline (Default)	0.500	0.762	0.0% (0.0–6.5%)	100.0%
No Waist (Primary)	Youden's J	0.082	0.762	78.2% (65.6–87.1%)	59.0%
No Waist (Primary)	Optimized (F2)	0.060	**0.762**	**90.9% (80.4–96.4%)**	48.7%
With Waist (Aux)	Baseline (Default)	0.500	0.763	0.0% (0.0–6.5%)	100.0%
With Waist (Aux)	Optimized (F2)	0.061	**0.763**	**89.1% (78.2–94.8%)**	49.0%

TABLE II-B. ESSENTIAL HYPERTENSION BENCHMARK (N=533)

Route	Model	Cutoff (τ)	AUC	Sens (95% CI)	Spec (%)
No Waist (Primary)	Baseline (Default)	0.500	0.757	50.3% (43.0–57.5%)	82.7%
No Waist (Primary)	Youden's J	0.339	0.757	71.3% (64.3–77.3%)	67.0%
No Waist (Primary)	Optimized (F2)	0.155	**0.757**	**90.6% (85.4–94.1%)**	40.9%
With Waist (Aux)	Baseline (Default)	0.500	0.755	49.7% (42.5–57.0%)	82.4%
With Waist (Aux)	Optimized (F2)	0.158	0.755	**88.4% (82.9–92.3%)**	41.8%

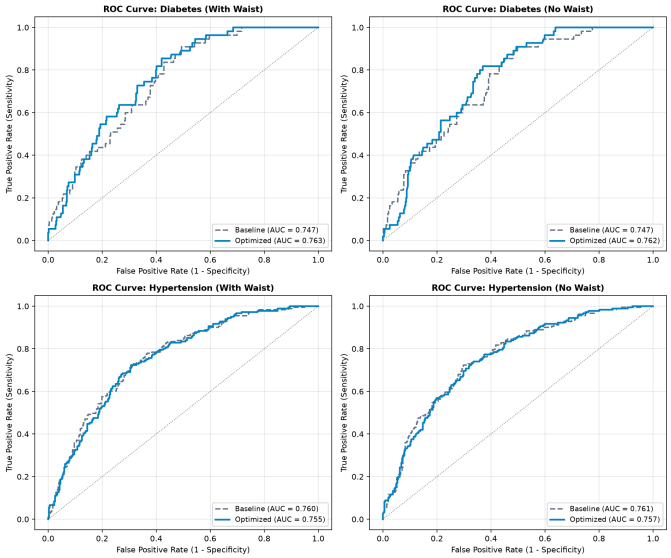


Fig. 2. Comparative ROC curves across clinical tasks on test splits.

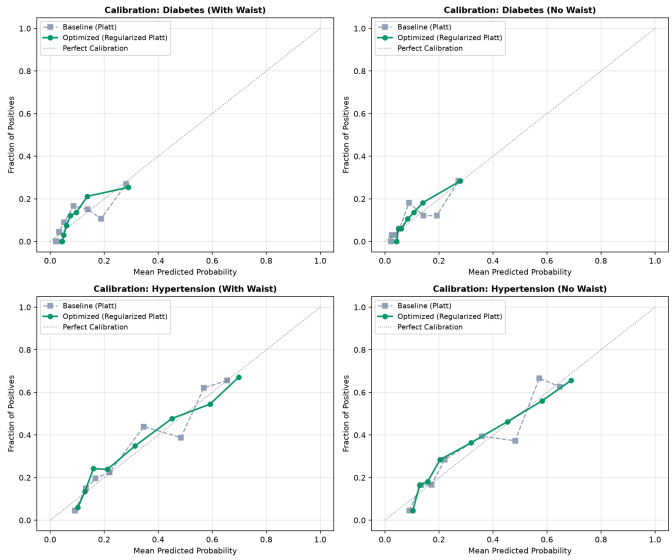


Fig. 3. Probability calibration reliability diagrams (quantile binned) confirming posterior probability mapping.

B. Quantitative Error Propagation Dynamics

To evaluate sensitivity under vision estimation noise, systematic BMI perturbation sweeps ($\Delta BMI \in [-2, +2]$ kg/m^2) were executed across the test cohort. Table III displays total body fat percentage shifts under both waist-inclusive and non-waist models.

TABLE III. BODY FAT SHIFT UNDER BMI PERTURBATION

Injected ΔBMI (kg/m^2)	Mean ΔBF With Waist	Attenuation (α)	Mean ΔBF No Waist
-2.0	-0.80 ± 0.80%	0.40	-2.02 ± 1.24%
-1.0	-0.37 ± 0.50%	0.37	-1.01 ± 0.82%
0.0	0.00 ± 0.00%	—	0.00 ± 0.00%
+1.0	+0.39 ± 0.54%	**0.39**	+1.00 ± 0.83%
+2.0	+0.77 ± 0.72%	0.38	+1.98 ± 1.14%

C. Clinical Triage Strata

The calibrated posterior probabilities are operationalized into four clinical triage strata configured in weights/thresholds.json:

1) Low Risk (Green): $p < 4.5\%$ (Diabetes), $p < 20.0\%$ (Hypertension). Annual wellness check.

2) Watchful (Yellow): $4.5\% \leq p < 6.0\%$ (Diabetes), $20.0\% \leq p < 33.0\%$ (Hypertension). Lifestyle counseling.

3) Screen Positive (Orange): $p \geq 6.0\%$ (Diabetes), $p \geq 33.0\%$ (Hypertension). Actionable trigger for confirmatory laboratory phlebotomy or clinical blood pressure cuff examination.

4) Urgent Risk (Red): $p \geq 12.0\%$ (Diabetes), $p \geq 55.0\%$ (Hypertension). Priority medical referral.

V. CLINICAL LIMITATIONS & ETHICAL BOUNDARIES

Face2Health is strictly an opportunistic triage tool, not a diagnostic instrument. Key limitations include: (1) target definitions rely on self-reported physician diagnoses (DIQ010/BPQ020) rather than active prospective laboratory testing; (2) potential domain shift between the US NHANES multi-ethnic training cohort and Asian/Southeast Asian populations regarding BMI-body fat relationships; (3) sensitivity to severe lighting extremes and head poses exceeding ±15° (intercepted by Face Guard); and (4) decoupled pipeline validation due to the absence of paired facial photographs in NHANES. Biometric data handling must comply with PDPA and GDPR guidelines, executing feature extraction locally on-device without storing raw facial images.

VI. CONCLUSION & FUTURE WORK

This study evaluated **Face2Health**, demonstrating that passive computer vision can be integrated into a calibrated triage instrument. By optimizing decision thresholds on validation

partitions, the primary non-invasive pipeline (without waist) achieved 90.91% screening sensitivity for Diabetes and 90.61% for Hypertension on held-out NHANES test data. Future work focuses on: (1) prospective clinical validation in Southeast Asian outpatient clinics; (2) integrating remote photoplethysmography (rPPG) for optical pulse wave velocity estimation; and (3) deploying lightweight ONNX models for offline mobile triage.

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